INTRA-FIRM TRADE AND PRODUCT CONTRACTIBILITY
(long version)

by

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CES 13-12 **March, 2013**

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Abstract

This paper examines the determinants of intra-firm trade in U.S. imports using detailed country-product data. We create a new measure of product contractibility based on the degree of intermediation in international trade for the product. We find important roles for the interaction of country and product characteristics in determining intra-firm trade shares. Intra- firm trade is high for products with low levels of contractibility sourced from countries with weak governance, for skill-intensive products from skill-scarce countries, and for capital-intensive products from capital-abundant countries.

Keywords: Related party trade, imports, contract theory, contractibility, intermediation, human capital, physical capital.

JEL classification: F23, F14, L14, L22, L23, L24

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† We thank Evan Gill, Justin Pierce and Jose Daniel Reyes for excellent research assistance, and the National Science Foundation for research support. Bernard thanks the European University Institute and Redding thanks the Centre for Economic Performance for research support. We thank Pol Antràs, Keith Head, Nathan Nunn, Emanuel Ornelas and conference seminar participants at the NBER and Paris for helpful comments. Empirical analysis was conducted at Census Research Data Centers. Any opinions and conclusions expressed herein are those of the authors and do not necessarily represent the views of the NSF, the NBER, or the U.S. Census Bureau. All results have been reviewed to ensure that no confidential information is disclosed.

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1. Introduction

Research on multinational firms has recently been extended to incorporate elements of contract theory. This literature addresses firms' decisions to source components in-house versus at arm's length and their choices over whether to locate production at home or abroad. It differs from earlier work on multinationals in its emphasis on the costs associated with writing contracts for specialized inputs and on the importance of traded intermediate goods.

This paper provides an empirical examination of the determinants of intra-firm trade. We use detailed U.S. import data to characterize the product and country attributes that determine firms' decisions to import from related parties rather than at arm's length. Theoretical models addressing this issue focus on the ability of the firm to write contracts for the production of specialized inputs. We introduce a new measure of products' revealed contractibility based on the idea that contracting is easier for products that are traded by intermediaries such as wholesalers.

Forty-six percent of U.S. imports occur between related parties in 2000. This aggregate statistic, however, obscures considerable variation in intra-firm intensity across import partners as well as products. Indeed, while 74 percent of U.S. imports from Japan are intra-firm, the figure for Bangladesh is just 2 percent. Likewise, trade between related parties accounts for 2 percent of U.S. imports of rubber and plastic footwear, but more than 70 percent of U.S. imports of autos, medical equipment and instruments. There is also significant variation in intra-firm intensity *across* countries *within* products.

These figures highlight the importance of product and country characteristics – and especially their interaction – in explaining intra-firm trade. Such factors are emphasized in recent theoretical models of multinational firms that stress the role of contracting in firms' decisions both to source components in-house versus at arm's length and to locate production at home versus abroad.[1] These models differ from earlier theories of multinationals in their emphasis on the costs associated with writing contracts for specialized inputs and the attention they pay to traded intermediate goods. Guided by these models, we examine the product and country determinants of intra-firm trade.

Our findings are related to the large theoretical literature on international trade and the boundaries of the firm, including in particular Antràs (2003), Antràs and Helpman (2004), and Grossman and Helpman (2002, 2005). Our findings are also related to the recent empirical literature examining the predictions of these models, including Corcos

[1] See, for example, Pol Antràs (2003), Pol Antràs and Elhanan Helpman (2004), and Gene M. Grossman and Elhanan Helpman (2005).

et al. (2008), Defever and Toubal (2007), Nunn and Trefler (2008) and Yeaple (2006). More generally, our findings are related to the recent literature on institutions and trade, including Levchenko (2007) and Nunn (2007).

We find that our measure of contractibility and countries' governance quality are associated with variation in intra-firm trade in interesting and intuitive ways, and that factors associated with engaging in related-party trade differ from those associated with the intensity of intra-firm trade once a link is established. Higher-quality country governance, for example, is associated with a higher probability of related-party trade taking place. Further increases in quality, however, coincide with *lower* shares of related-party trade, presumably due to the greater ease with which arm's-length contracts can be written. With respect to interactions of product and country attributes, improvements in country governance lead to the largest reductions in intra-firm trade in low contractibility products.

2. Data

We use the U.S. Linked/Longitudinal Firm Trade Transaction Database (LFTTD), which links individual U.S. trade transactions to U.S. firms.[2] For each import transaction, we observe the U.S. firm engaging in the transaction, the ten-digit Harmonized System (HS) classification of the product shipped, the (nominal) value shipped, the shipment date, the source country, and whether the transaction takes place at "arm's length" (AL) or between "related parties" (RP). Import partners are "related" if either party owns, directly or indirectly, 6 percent or more of the other party.[3] To concord SIC production and HS trade data, and to expand the sample of countries on which data on country characteristics are available, we focus on the year 1997.

To explore the role of various country characteristics discussed below, we combine these trade data with measures of physical capital abundance, human capital abundance, and population from Robert E. Hall and Charles I. Jones (1999), a composite index of countries' governance quality from the World Bank, and measures of trade and FDI protection from Heritage Foundation/WSJ (2006). We use factor analysis to create a univariate measure of country governance for 1996 from the six World Bank measures reported by Daniel Kaufman, Aart Kraay and Massimo Mastruzzi (2006). The first factor accounts for around 90 percent of the variance of each of the six component measures and

[2] See Andrew B. Bernard, J. Bradford Jensen and Peter K. Schott (2009) for more details.

[3] This dataset excludes the U.S. Postal Service and firms in agriculture, forestry and fishing, railroads, education, public administration and several smaller sectors.

we use this factor as the indicator of governance in our empirical work.

We measure products' capital and skill intensity using data from the 1997 U.S. Census of Manufactures. We assign all ten-digit HS products within a particular four-digit SIC industry the average physical capital or skill intensity of all plants whose output is concentrated in that industry. Physical capital intensity is measured as the log of the book value of plant and equipment per employee while skill is non-production workers as a share of employment. Industry headquarters intensity is measured by the average share of firm employment at headquarters and auxiliary establishments.

3. Intra-firm imports

This section documents the extent of U.S. intra-firm imports by trading partner and industry. To maximize our ability to report results across countries and industries, we use recently published, publicly available data on related-party trade from the Foreign Trade Division of the U.S. Census Bureau.[4] The industry data on related-party trade is reported according to the North American Industry Classification System (NAICS) and, as a result, differs from the more detailed Harmonized System codes available in the LFFTD and employed in the subsequent regression analysis.

3.1. By Country

We begin by considering variation in related-party imports across countries in 2000. The data are summarized in Table 1 which reports the level of imports and the share of related-party imports by country. Over 46 percent of U.S. imports are intra-firm and there is wide range in intensity of intra-firm trade across countries. For the average country, 23.8 percent of exports to the U.S. are intra-firm and more than a quarter of countries have intra-firm shares less than 5 percent. On the low end, imports from Bangladesh are almost entirely arms-length transactions, with just 2 percent of the total value of imports taking place inside the firm. In contrast, imports from Japan and Ireland are dominated by intra-firm transactions. In 2000, 76 percent of the value of imports from Ireland and 74 percent of the imports from Japan were conducted by multinationals trading with related foreign divisions. Anecdotal publicly-available evidence would suggest that the intra-firm imports of Ireland and Japan stem from different types of organizations. Japanese intra-firm shipments to the U.S. are likely trades between Japanese parents and

[4]We choose 2000 as it is the year closest to the product-country import data used in our empirical specifications below. The original data source for all the results in this section is http://sasweb.ssd.census.gov/relatedparty.

U.S. subsidiaries, while Irish intra-firm shipments are more likely to be between Irish subsidiaries and U.S. parents, or U.S. affiliates of European multinationals.

In general, countries that account for low shares of U.S. intra-firm imports are less developed and have lower overall import volumes, while high-income countries in the OECD generally report above average intra-firm imports to the United States. Imports from China, the fourth largest source of U.S. imports in 2000, are still largely conducted between unrelated parties with just 18 percent exchanged inside the firm.

3.2. By Industry

As with the country-level data, industries vary widely in the extent to which their trade takes place within firms.[5] Imports of leather, textiles and apparel are dominated by arms-length transactions while more than half of imports in transportation equipment, computers and electronics products and chemicals are conducted between related parties. Table 2 reports the manufacturing industries with the 20 highest and 20 lowest shares of related-party trade in 2000 using 6-digit NAICS industries. Footwear industries are heavily represented in the low end of the distribution of intra-firm trade shares. In rubber and plastic footwear, for example, intra-firm imports account for just 1.8 percent of total imports. Imports of autos and related equipment, medical equipment and pharmaceuticals, and instruments, on the other hand, are dominated by intra-firm transactions. In each of these industries, more than 70 percent of all imports are between related parties.

These industry averages obscure important variation across countries within products. Figure 1 shows the distribution of imports of Photo Films, Plates and Chemicals (NAICS 325992) across countries. This industry has fifth highest share of intra-firm imports. The figure shows both the share of intra-firm imports from each country (line - left axis) and the level of overall imports (bar - log scale right axis). The countries are sorted by the share of intra-firm imports in total imports in 2000. While the industry as a whole has a high level of intra-firm trade, there is substantial variation across countries. Half the countries, including most of the major exporters by volume, have intra-firm shares greater than 70 percent. Most of the remaining countries, including a number of middle income and developing countries, have little or no related-party trade to the U.S..

This pattern of heterogeneous intra-firm shares across countries within industries is the norm rather than the exception. Figure 2 shows the same picture for imports of Other

[5] In this section we use publicly available data from the foreign trade division of the Census Bureau. As a consequence these table use the NAICS industry classification system. In our regression results below we use the much more disaggregated 10-digit products of the Harmonized System.

Footwear (NAICS 316219).[6] This industry has sixth lowest share of intra-firm imports but again there is a wide variation in related party shares ranging from 100 percent to zero. This variation in intra-firm imports shares within industries across countries motivates our use of both country and product characteristics and their interaction in our subsequent empirical work.

4. "Revealed Contractibility"

We assume that products passing through intermediaries are the easiest over which to contract. As a result, we measure products' "revealed" contractibility as the weighted average wholesale employment share of firms importing the product, using firms' import value as weights,

$$IMED_p = \sum_f \frac{W_f}{EMP_f} \frac{M_{pf}}{M_p}. \quad (1)$$

The first term in the intermediation measure is the share of wholesale employment (W_f) in firm f's total employment (EMP_f).[7] The second term is the import share of firm f in ten-digit HS product market p, with M_{pf} and M_p representing firm f's imports of product p and total U.S. imports of product p, respectively. Intermediation ranges between zero and unity: if no firms importing product p have any wholesale establishments, $IMED_p = 0$. On the other hand, if product p is imported exclusively by firms with 100 percent employment in wholesaling, $IMED_p = 1$.

Table 3 reports the intermediation measure for HS2 industries in 1997. Industries are sorted according to intermediation, from low to high. Across industries, intermediation averages 0.241, ranging from 0.012 in non-railway vehicles (HS 87) to 0.631 in lead (HS 78), with an interquartile range of 0.123 to 0.345. Agricultural goods and relatively labor intensive industries such as apparel and footwear generally have the highest measured intermediation, while more "sophisticated" products such as vehicles, pharmaceuticals, chemicals and photographic goods have the lowest measures of intermediation.

Intermediation and intra-firm import shares are inversely related across two-digit HS categories, as shown in Figure 1. There is however substantial independent variation in the two variables, as industries with similar levels of intermediation span a wide range of intra-firm intensity. Footwear (HS 64) and Organic Chemicals (HS 29), for example, have

[6] Only countries with more than $100,000 of of U.S. imports are shown.

[7] We observe employment at the establishment level and therefore assign all employees in an establishment to the major industry of the establishment. Firms with a single establishment necessarily have 100 percent employment in a single industry. Wholesale is NAICS sector 42.

comparable levels of intermediation, 0.135 and 0.136 respectively. However, more than half of Organic Chemicals imports are conducted by related parties while the intra-firm trade share for Footwear is less than 10 percent.

5. Determinants of intra-firm trade

Our empirical analysis uses cross-sectional data on intra-firm and total U.S. imports of product p from county c in 1997. Our empirical specification regresses measures of intra-firm trade (IF_{pc}) on product characteristics (X_p), country characteristics (Z_c) and interactions between product and country characteristics ($X_p \times Z_c$):

$$IF_{pc} = \theta + \alpha X_p + \beta Z_c + \gamma (X_p Z_c) + \epsilon_{pc}, \tag{2}$$

We consider two measures of intra-firm trade: the share of intra-firm imports in U.S. imports, which we refer to as the "intensive" margin, and a dummy variable which is equal to one if there are positive intra-firm imports for a product from a country, which we call the "extensive" margin. In constructing the interaction terms, we subtract the sample mean from each variable entering the interaction term. This normalization ensures that the main effects of each variable can be interpreted as the effect at the sample mean.

Our choice of product and country characteristics is motivated by the recent theoretical literature on contractual frictions and international trade. This literature emphasizes the relative importance of relationship-specific investments by headquarters and supplier firms and the degree of verifiability of these investments. In Antràs (2003), capital intensity captures the relative importance of headquarters' investments, and hence we include industry capital intensity and country capital abundance. To allow for the possibility that other factor intensities matter, we also include industry skill intensity and country skill abundance. In Antràs and Helpman (2004), headquarters investments are interpreted more broadly, and hence we include the direct measure of headquarters intensity noted above. In Grossman and Helpman (2005), the degree of verifiability of relationship-specific investments can vary with product and country characteristics, and hence we include revealed product contractibility and country governance as further independent variables. Finally, we explore the impact of policy-based barriers by including measures of trade and FDI protection as country characteristics.

Table 4 reports the results of estimating specification (2). Columns (1) and (3) use the extensive margin as the dependent variable, so the sample comprises all product-country cells with positive imports, including those with zero intra-firm trade. Columns (2) and

(4) focus on the intensive margin, and the sample is all observations with positive intra-firm trade. Columns (3) and (4) control for the non-random selection of observations with positive intra-firm imports using the Heckman two-stage estimation procedure. The two stages are separately identified by functional form and the excluded variable from the second-stage regression. For the excluded variable, we choose the cost of phone calls to the US, which arguably affects the fixed costs of establishing an affiliate but not the relative variable costs of intra-firm versus arms-length trade.[8]

We find an important role for revealed contractibility on both the intensive and extensive margins of intra-firm trade. Consistent with the recent theoretical literature on contractual frictions in international trade, columns (1) and (2) show that higher revealed product contractibility is associated with less intra-firm trade. The role of the contracting environment varies across the intensive and extensive margins. Increases in governance quality raise the probability that foreign affiliates are present (column 1), but are associated with lower shares of intra-firm trade (column 2). This result suggests good governance promotes the establishment of related-party trade but not its intensity once established, which is consistent with the idea that arm's-length contracting is easier in countries with good governance. This non-linearity in the role of the country contracting environment is not formally developed in existing theoretical models. Similar differences between the intensive and extensive margins are present for population and FDI protection.

Results in Table 4 also indicate the significance of interactions of product and country characteristics in determining intra-firm trade. While the main effects for intermediation and country governance are both negative in column (4), the interaction term has a positive coefficient. That is, improved governance is associated with less intra-firm trade, especially for goods with lower revealed contractibility.

In contrast to previous work, we also find a role for industry skill intensity and country skill abundance. The main effects of industry skill intensity on intra-firm trade are positive for both the intensive and extensive margins; the main effects of country human capital abundance are negative; and the estimated coefficients on the skill interaction terms are negative. Therefore, greater industry skill intensity increases the share of intra-firm trade, and leads to larger increases in more skill-scarce countries. In contrast, greater country skill abundance reduces the share of intra-firm trade, and leads to larger reductions in more skill-intensive products. As in Antràs (2003), industry capital intensity

[8]The likelihood ratio test of rho=0 yields a chi-squared statistic of 26.21, rejecting the null of independent equations.

and country capital abundance play a role in determining the share of intra-firm trade. The positive coefficient on the interaction between industry capital intensity and country capital abundance implies that intra-firm trade shares are high for capital-intensive products coming from capital-abundant countries. Both FDI and trade protection influence intra-firm trade; headquarters intensity is not statistically significantly associated with intra-firm trade shares.

In Table 5, we repeat the OLS specification from column (2) in Table 4 with a complete set of country and product fixed effects to examine the robustness of the results on the interaction terms. The contractibility-governance and human capital interactions retain their sign and significance while the interaction on physical capital is insignificant.

5.1. Robustness

In this section we explore the robustness of the results to alternative specifications. Column 1 of Table 6 repeats the preferred specification from column 4 in Table 4. In columns 2-4, we drop sectors that contain firms that do not conform strictly to the existing theoretical models. The literature on multinationals and contracting envisions a producing firm headquartered in an advanced country importing intermediate goods, potentially from its affiliates. Our results in column 1 include all imports, including imports of final goods and imports by U.S. affiliates of foreign multinationals. Column 2 excludes sectors that are intensive in foreign-owned firms, column 3 drops final goods products and column 4 drops both at once.[9] None of the coefficients change sign or significance and all the main conclusions are robust to these sample changes.

In the final three columns of Table 6, we include additional regressors considered in related empirical work. Column 5 adds a measure of industry R&D intensity, the R&D to sales ratio which is only available for a subset of industries.[10] The R&D coefficient is positive and significant, confirming results in Antràs (2003), Yeaple (2006) and others. Adding industry R&D intensity eliminates the significance of the physical capital interaction as well as that of human capital intensity. Finally in columns 6 and 7, we add the measure of contractibility suggested by Nunn (2007) based on the proportion of each

[9]To identify sectors that are intensive in foreign affiliate imports, we use the Bureau of Economic Analysis measure of US imports shipped to affiliates by the foreign parent group by sector. We construct a measure foreign input intensity by dividing the imports shipped to affiliates by employment in an industry. High foreign affiliate industries are those above the mean. Data is available at http://www.bea.gov/scb/account_articles/international/iidguide.htm#FDIUS. We follow the classification of Sitchinava (2007) to identify product categories that are final good imports. All columns of Table 6 report the second stage of a Heckman specification with the cost of phone calls as the excluded variable in the second stage.

[10]R&D are available from the NSF at http://www.nsf.gov/statistics/iris/history_pub.cfm.

industry's intermediate inputs that are relationship-specific and therefore susceptible to potential contracting problems.[11] Column 7 includes an interaction of the Nunn measure with the country governance measure. The Nunn measure is positive and significant, as found by Nunn and Trefler (2008), but does not affect the revealed contractibility measure or its interaction. The interaction term is negative and significant, which combined with the negative main effect of country governance implies that improvements in country governance are associated with the largest reductions in intra-firm trade in sectors with more relationship-specific inputs. These results suggest that the Nunn measure of input sophistication and our measure of intermediation may be capturing different aspects of product contractibility both of which interact with country governance in shaping whether trade occurs within the boundary of the firm.

6. Conclusions

The literature on firms and international trade has focused attention on issues of contracting and the boundaries of the firm. This research speaks to policy issues surrounding the growth of outsourcing, offshoring and international production networks.

Our results provide evidence on the role of country governance and product contractibility in determining intra-firm trade. We find evidence of selection: the decision to establish a foreign affiliate in a country differs from the choice of how much to source from the affiliate once it is established. While affiliates are more likely to be situated in countries that are larger and have better governance, once affiliates exist, the share of intra-firm trade is negatively related to both country size and country governance quality.

Our findings both complement and extend the existing empirical literature on intra-firm trade. Our results confirm the role of industry capital intensity and country capital abundance in influencing intra-firm trade. Our results also point to the role of other interactions between country and product characteristics and their interactions.

[11] According to Nunn (2007), relationship specific inputs are those that are not traded on organized exchanges as measured by Rauch (1999).

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Country/Terrority	Total Imports (millions)	Related-Party Import Share
Brunei	387	0.000
Lesotho	140	0.000
Equatorial Guinea	155	0.000
Palau	14	0.001
Turkmenistan	28	0.001
Micronesia	14	0.002
Republic of Yemen	151	0.002
Mozambique	24	0.003
Botswana	41	0.003
Swaziland	53	0.005
Oman	257	0.006
Uzbekistan	35	0.007
Mauritius	286	0.008
Algeria	2,690	0.008
Cambodia	824	0.009
Faroe Islands	31	0.011
Madagascar	158	0.011
Namibia	42	0.015
Bermuda	39	0.015
Bangladesh	2,416	0.019
Ethiopia	29	0.023
Burma (Myanmar)	468	0.024
Macedonia (Skopje)	138	0.025
Cameroon	146	0.027
Uganda	29	0.031
Peru	1,985	0.032
British Virgin Islands	31	0.032
Nepal	229	0.032
Uruguay	309	0.033
Pakistan	2,164	0.035
Haiti	297	0.035
Macao	1,265	0.036
Fiji	146	0.036
Angola	3,343	0.036
Estonia	542	0.038
Mongolia	117	0.039
Iran	169	0.043
Paraguay	42	0.045
Papua New Guinea	37	0.046
Jordan	73	0.046
Azerbaijan	20	0.046
Lebanon	76	0.050
Vietnam	827	0.053
Moldova	105	0.060
Armenia	23	0.063
Zimbabwe	113	0.065
Greenland	16	0.066
Grenada	27	0.066
Syria	150	0.068
Latvia	295	0.068
Egypt	925	0.070
Belize	91	0.073
United Arab Emirates	937	0.073
India	10,680	0.077
Nicaragua	597	0.078
Qatar	491	0.078
Bulgaria	231	0.081
Guyana	127	0.083
Belarus	104	0.086
Cyprus	23	0.086
Ecuador	2,267	0.089
Turkey	3,027	0.095
Kenya	109	0.097
Panama	297	0.105
Ghana	206	0.114
Guatemala	2,603	0.122
Lithuania	132	0.123
Netherlands Antilles	721	0.126
Ivory Coast	367	0.136
Sri Lanka	2,002	0.137
Hong Kong	11,350	0.140
Tanzania	35	0.142
Congo (Kinshasa)	212	0.145
Barbados	38	0.145
South Africa	4,204	0.151
Venezuela	17,430	0.151
Greece	602	0.157
Georgia	24	0.160
Argentina	3,095	0.163
Bahrain	338	0.164
Ukraine	872	0.166
Zambia	18	0.173
Chile	3,258	0.179
China	99,580	0.181
Malawi	68	0.189
Tunisia	91	0.200
Romania	471	0.215
Taiwan	40,380	0.216
Croatia	141	0.218
Indonesia	10,320	0.228
Colombia	6,681	0.228
Aruba	1,222	0.229
French Polynesia	44	0.231
Spain	5,674	0.241
Slovenia	314	0.242
Israel	12,950	0.248
Kazakhstan	432	0.253
Trinidad and Tobago	2,179	0.253
Russia	7,761	0.266
Congo (Brazzaville)	508	0.272
Poland	1,040	0.275
Monaco	23	0.275
Australia	6,213	0.290
Brazil	13,730	0.293
Bosnia-Hercegovina	18	0.296
Kuwait	2,499	0.307
Italy	24,790	0.312
Portugal	1,571	0.321
Bolivia	184	0.327
Czech Republic	1,069	0.334
Thailand	16,300	0.340
St Lucia	22	0.342
Norway	5,540	0.353
Nigeria	9,680	0.364
Maldives	94	0.368
Iraq	4,393	0.372
New Zealand	2,055	0.379
Morocco	456	0.380
Gabon	2,038	0.403
France	29,430	0.410
Belgium	9,844	0.415
Iceland	260	0.416
Slovakia	242	0.421
Bahamas	273	0.434
Canada	229,100	0.440
Denmark	2,953	0.451
El Salvador	1,925	0.456
Dominican Republic	4,378	0.459
Jamaica	632	0.475
United Kingdom	42,840	0.488
St Kitts and Nevis	37	0.493
Philippines	13,940	0.496
Austria	3,118	0.506
Honduras	3,091	0.519
Suriname	135	0.523
Switzerland	10,090	0.536
Netherlands	9,679	0.536
Saudi Arabia	14,330	0.549
Korea, South	39,830	0.554
Luxembourg	331	0.575
Finland	3,238	0.617
Malaysia	25,450	0.645
Germany	58,350	0.647
Mexico	134,700	0.661
Malta	462	0.675
Costa Rica	3,555	0.692
Hungary	2,711	0.694
Sweden	9,570	0.700
Singapore	19,110	0.727
Japan	145,700	0.743
Ireland	16,370	0.761
Guinea	88	0.882
Liechtenstein	293	0.886
Liberia	45	0.888
New Caledonia	31	0.972

Table 1: U.S. Imports and Related-Party Share By Country, 2000

20 Lowest Related-party Import Shares (NAICS 6-digit)	Total Imports (millions$)	Related-Party Imports (millions$)	Related-Party Import Share
Motor Homes	119	1	0.004
Rubber & Plastic Footwear	584	10	0.018
Primary Smelting & Refining of Copper	2,396	66	0.027
Missile/Space Veh Parts & Auziliary Equip, NESOI	224	7	0.033
Cut Stone & Stone Products	1,281	44	0.034
Other Footwear	4,164	151	0.036
Folding Paperboard Boxes	385	16	0.041
Jewelers' Material & Lapidary Work	13,228	582	0.044
Canvas & Related Products	234	11	0.048
Prefabricated Wood Buildings	104	6	0.053
Dried and Dehydrated Foods	161	9	0.056
Spices & Extracts	501	29	0.058
Women's Footwear (Exc Athletic)	6,012	349	0.058
Women's/Girl's Dresses	2,104	126	0.060
Fur & Leather Apparel	1,973	121	0.061
Men's Footwear (Exc Athletic)	3,590	230	0.064
Hats & Caps	923	63	0.068
Wines	2,706	204	0.075
Softwood Veneer & Plywood	271	21	0.077
Miscellaneous Wood Products	1,765	140	0.079
20 Highest Related-Party Import Shares (NAICS 6-digit)			
Prepared Flour Mixes & Dough	123	89	0.722
Electromedical Apparatus	3,129	2,262	0.723
Automatic Environmental Controls	619	450	0.727
Motor Vehicle Gasoline engines & Engine Parts	10,262	7,504	0.731
Sanitary Paper Products	736	538	0.731
Telephone Apparatus	13,041	9,552	0.732
Motor Vehicle Electrical & Electronic Equip, Nesoi	7,337	5,374	0.732
Medicinal & Botonical Drugs & Vitamins	17,400	12,823	0.737
Carbon Paper & Inked Ribbon	314	233	0.741
Pharmaceutical Preparations	10,131	7,591	0.749
Motor Vehicle Air-Conditioning	1,225	919	0.750
Bottled Water	200	151	0.755
Tires & Tire Parts (Excl Retreadings)	4,720	3,587	0.760
Computer Storage Devices	16,283	12,683	0.779
Pesticides & Other Agricultural Chemicals	500	401	0.802
Photo Films, Papers, Plates & Chemicals	2,485	2,026	0.815

Table 2: U.S. Related Party Trade by 6-Digiti NAICS Industry, 2000

Chapter	Description	Intermediation	Chapter	Description	Intermediation
87	Non-Railway vehicles	0.012	51	Wool, woven fabric	0.223
27	Mineral fuels, oils, waxes	0.019	62	Apparel, not knitted or crochete	0.232
1	Live animals	0.023	33	Oils; perfumery	0.234
88	Aircraft, spacecraft	0.024	22	Beverages, spirits	0.241
86	Railway locomotives	0.025	79	Zinc and articles thereof	0.242
30	Pharmaceutical products	0.027	69	Ceramic products	0.247
26	Ores, slag and ash	0.030	36	Explosives	0.247
89	Ships, boats, etc.	0.034	96	Misc. manufactured articles	0.259
37	Photographic goods	0.043	21	Misc. edible preparations	0.262
75	Nickel and articles thereof	0.050	80	Tin and articles thereof	0.274
31	Fertilisers	0.056	54	Man-made filaments	0.282
97	Works of art, antiques	0.068	63	Other made up textile articles	0.291
85	Electrical machinery	0.084	56	Wadding, yarns, ropes, cables	0.293
47	Pulp of wood	0.088	68	Stone, plaster, cement	0.295
38	Misc. chemical products	0.090	15	Animal, vegetable fats and oils	0.297
28	Inorganic chemicals+Z77	0.094	11	Milling industry products	0.301
90	Instruments	0.095	42	Leather; saddlery and harness	0.314
76	Aluminum and articles thereof	0.100	91	Clocks and watches	0.322
48	Paper; articles of paper pulp	0.101	50	Silk	0.327
84	Nuclear reactors, machinery	0.102	92	Musical instruments	0.327
25	Salt; earths and stone	0.106	66	Umbrella, walking-sticks	0.334
24	Tobacco	0.108	16	Preparations of meat, fish	0.339
40	Rubber and articles thereof	0.118	2	Meat	0.341
49	Printed books, newspapers	0.122	8	Fruit and nuts	0.345
17	Sugars	0.123	41	Raw hides, skins, leather	0.345
23	Residues from food industries	0.130	58	Woven fabrics; tapestries	0.369
71	Pearls, precious metals, coin	0.135	93	Arms and ammunition	0.373
29	Organic chemicals	0.135	55	Man-made staple fibres	0.373
64	Footwear, gaiters	0.136	13	Gums, resins	0.374
70	Glass and glassware	0.141	46	Straw; basketware	0.379
32	Tanning or dyeing extracts	0.162	57	Carpets, floor coverings	0.384
35	Starches, glues, enzymes	0.168	45	Cork articles	0.409
10	Cereals	0.172	14	Vegetable products	0.414
81	Other base metals	0.173	65	Headgear and parts thereof	0.415
18	Cocoa	0.175	5	Animal products	0.433
94	Furniture; prefab buildings	0.179	4	Dairy produce; honey	0.433
12	Oil seeds, grains, plants	0.181	67	Feathers and down articles	0.440
74	Copper and articles thereof	0.186	20	Vegetables, fruit, nuts	0.447
39	Plastics and articles thereof	0.192	43	Furskins and artificial fur	0.454
72	Iron and steel	0.193	7	Vegetables	0.460
34	Soap, waxes, candles	0.195	60	Knitted or crocheted fabrics	0.465
83	Misc. articles of base metal	0.196	3	Fish, crustaceans	0.469
95	Toys, games	0.199	9	Coffee, tea, spices	0.474
61	Knitted or crocheted apparel	0.207	19	Cereals, flour, milk	0.482
59	Textile fabrics	0.211	53	Vegetable textile fibres	0.500
82	Tools, implements, cutlery	0.212	6	Trees and plants	0.516
44	Wood articles; wood charcoal	0.213	52	Cotton	0.529
73	Articles of iron or steel	0.214	78	Lead and articles thereof	0.631

Table 3: Intermediation Index by HS2 Industry, 1997

	(1) Intra-Firm Trade Dummy	(2) Share of Intra-Firm Trade	(3) Intra-Firm Trade Dummy	(4) Share of Intra-Firm Trade
Intermediation	-0.715 ***	-0.165 ***	-0.719 ***	-0.235 ***
	(0.050)	(0.019)	(0.050)	(0.025)
Governance	0.154 ***	-0.031 ***	0.103 ***	-0.031 ***
	(0.014)	(0.007)	(0.019)	(0.009)
x Intermediation	-0.058	0.084 ***	-0.056	0.090 ***
	(0.039)	(0.015)	(0.039)	(0.017)
Capital Intensity	-0.005	0.059 ***	-0.005	0.056 ***
	(0.021)	(0.007)	(0.020)	(0.008)
Log capital abundance	0.213 ***	0.067 ***	0.173 ***	0.068 ***
	(0.016)	(0.006)	(0.017)	(0.007)
x Capital intensity	0.068 ***	0.005	0.072 ***	0.010 **
	(0.016)	(0.004)	(0.015)	(0.005)
Skill Intensity	1.336 ***	0.196 ***	1.348 ***	0.324 ***
	(0.192)	(0.051)	(0.192)	(0.067)
Log human capital abundance	-0.105 **	-0.066 ***	-0.044	-0.059 **
	(0.044)	(0.022)	(0.046)	(0.023)
x Skill intensity	-0.415	-1.063 ***	-0.460	-1.142 ***
	(0.407)	(0.152)	(0.411)	(0.174)
HQ Intensity	-0.103	0.043	-0.099	0.016
	(0.196)	(0.065)	(0.196)	(0.071)
Log population	0.152 ***	-0.034 ***	0.145 ***	-0.033 ***
	(0.008)	(0.002)	(0.009)	(0.003)
FDI protection	0.13 ***	-0.017 ***	0.154 ***	0.039 ***
	(0.015)	(0.005)	(0.014)	(0.007)
Trade protection	-0.098 ***	0.017 ***	-0.092 ***	-0.023 ***
	(0.011)	(0.004)	(0.011)	(0.005)
US Phone Call Cost	-	-	-0.050 ***	-
			(0.003)	
Lambda	-	-	-	0.150 ***
				(0.029)
Sample	Full	Positive Intra-firm Trade	Full	Positive Intra-firm Trade
Estimation	Probit	OLS	Heckman First-Stage	Heckman Second-Stage
R-squared		0.079		
Observations	180,774	92,656	180,774	92,656

Note: In constructing the interaction terms, we subtract the sample mean from each variable entering the interaction term, so that the main effects of each variable can be interpreted as the effect at the sample mean. Columns 1 and 3 include all country-product pairs with positive imports. Robust standard errors adjusted for clustering at the four-digit SIC level are reported below coefficient estimates. ***, **, and * indicate significance at the 1, 5, and 10 percent levels respectively.

Table 4: Determinants of Intra-Firm Imports, HS10-Country 1997

	(1)	(2)	(3)	(4)
Intermediation	-0.165 ***	-0.156 ***		
	(0.019)	(0.019)		
Governance	-0.031 ***		-0.033 ***	
	(0.007)		(0.008)	
x Intermediation	0.084 ***	0.074 ***	0.097 ***	0.086 ***
	(0.015)	(0.016)	(0.016)	(0.016)
Capital Intensity	0.059 ***	0.055 ***		
	(0.007)	(0.007)		
Log capital abundance	0.067 ***		0.052 ***	
	(0.006)		(0.006)	
x Capital intensity	0.005	-0.003	-0.001	-0.007
	(0.004)	(0.006)	(0.005)	(0.007)
Skill Intensity	0.196 ***	0.192 ***		
	(0.051)	(0.051)		
Log human capital abundance	-0.066 ***		-0.055 ***	
	(0.022)		(0.021)	
x Skill intensity	-1.063 ***	-1.297 ***	-1.175 ***	-1.45 ***
	(0.152)	(0.181)	(0.155)	(0.192)
HQ Intensity	0.043	0.068		
	(0.065)	(0.061)		
Log population	-0.034 ***		-0.045 ***	
	(0.002)		(0.002)	
FDI protection	-0.017 ***		-0.017 ***	
	(0.005)		(0.005)	
Trade protection	0.017 ***		0.021 ***	
	(0.004)		(0.004)	
Estimation	OLS	OLS	OLS	OLS
Sample	Positive Intra-firm Trade	Positive Intra-firm Trade	Positive Intra-firm Trade	Positive Intra-firm Trade
Fixed Effects	None	Country	Product	Country and Product
Observations	92,656	92656	92656	92656
R-squared	0.079	0.154	0.300	0.368

Note: Column (1) repeats the results from column (2) in Table 5. In constructing the interaction terms, we subtract the sample mean from each variable entering the interaction term, so that the main effects of each variable can be interpreted as the effect at the sample mean. Robust standard errors adjusted for clustering at the four-digit SIC level are reported below coefficient estimates. ***, **, and * indicate significance at the 1, 5, and 10 percent levels respectively.

Table 5: Determinants of Intra-Firm Imports - Fixed Effects, HS10-Country 1997

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Intermediation	-0.235 ***	-0.214 ***	-0.235 ***	-0.208 ***	-0.243 ***	-0.212 ***	-0.206 ***
	(0.025)	(0.027)	(0.027)	(0.026)	(0.028)	(0.023)	(0.023)
Governance	-0.031 ***	-0.023 ***	-0.048 ***	-0.043 ***	-0.059 ***	-0.031 ***	-0.029 ***
	(0.009)	(0.010)	(0.008)	(0.008)	(0.008)	(0.009)	(0.009)
x Intermediation	0.09 ***	0.069 ***	0.107 ***	0.094 ***	0.107 ***	0.089 ***	0.066 ***
	(0.017)	(0.017)	(0.014)	(0.014)	(0.019)	(0.017)	(0.015)
Capital Intensity	0.056 ***	0.06 ***	0.063 ***	0.074 ***	0.079 ***	0.079 ***	0.085 ***
	(0.008)	(0.009)	(0.012)	(0.013)	(0.018)	(0.010)	(0.010)
Log capital abundance	0.068 ***	0.073 ***	0.043 ***	0.042 ***	0.068 ***	0.068 ***	0.07 ***
	(0.007)	(0.009)	(0.008)	(0.010)	(0.007)	(0.007)	(0.006)
x Capital intensity	0.01 ***	0.011 ***	0.027 ***	0.025 ***	0.006	0.01 ***	-0.004
	(0.005)	(0.004)	(0.005)	(0.005)	(0.008)	(0.005)	(0.004)
Skill Intensity	0.324 ***	0.315 ***	0.325 ***	0.372 ***	0.084	0.198 ***	0.188 ***
	(0.067)	(0.084)	(0.063)	(0.073)	(0.077)	(0.070)	(0.071)
Log human capital abundance	-0.033	-0.034 ***	-0.034 ***	-0.037 ***	-0.034 ***	-0.033 ***	-0.033 ***
	(0.003)	(0.004)	(0.004)	(0.004)	(0.003)	(0.003)	(0.003)
x Skill intensity	-1.142 ***	-0.968 ***	-0.804 ***	-0.718 ***	-0.982 ***	-1.141 ***	-0.818 ***
	(0.174)	(0.192)	(0.158)	(0.182)	(0.168)	(0.174)	(0.201)
HQ Intensity	0.016	0.061	-0.078	-0.049	-0.16 *	0.011	0
	(0.071)	(0.076)	(0.109)	(0.125)	(0.093)	(0.072)	(0.071)
Log population	-0.059 ***	-0.066 ***	-0.066 ***	-0.075 ***	-0.089 ***	-0.059 ***	-0.068 ***
	(0.023)	(0.025)	(0.026)	(0.030)	(0.031)	(0.023)	(0.023)
FDI protection	0.039 ***	0.038 ***	0.031 ***	0.029 ***	0.041 ***	0.039 ***	0.039 ***
	(0.007)	(0.008)	(0.008)	(0.009)	(0.007)	(0.007)	(0.007)
Trade protection	-0.023 ***	-0.02 ***	-0.037 ***	-0.037 ***	-0.027 ***	-0.023 ***	-0.024 ***
	(0.005)	(0.005)	(0.003)	(0.004)	(0.005)	(0.005)	(0.005)
R&D Intensity					1.230 ***		
					(0.245)		
Nunn measure						0.143 ***	0.185 ***
						(0.044)	(0.044)
x Governance							-0.141 ***
							(0.021)
Lambda	0.150 ***	0.164 ***	0.093 ***	0.085 ***	0.112 ***	0.144 ***	0.150 ***
Sample	Full	Exclude foreign-intensive industries	Exclude final goods	Exclude both foreign-intensive and final goods	Industries with R&D data	Full	Full
Observations	180774	163005	131424	115546	112337	179790	179790

Note: Column (1) replicates the baseline results from column (4) in Table 5. In constructing the interaction terms, we subtract the sample mean from each variable entering the interaction term, so that the main effects of each variable can be interpreted as the effect at the sample mean. Robust standard errors adjusted for clustering at the four-digit SIC level are reported below coefficient estimates. ***, **, and * indicate significance at the 1, 5, and 10 percent levels respectively.

Table 6: Determinants of Intra-Firm Imports - Robustness, HS10-Country 1997

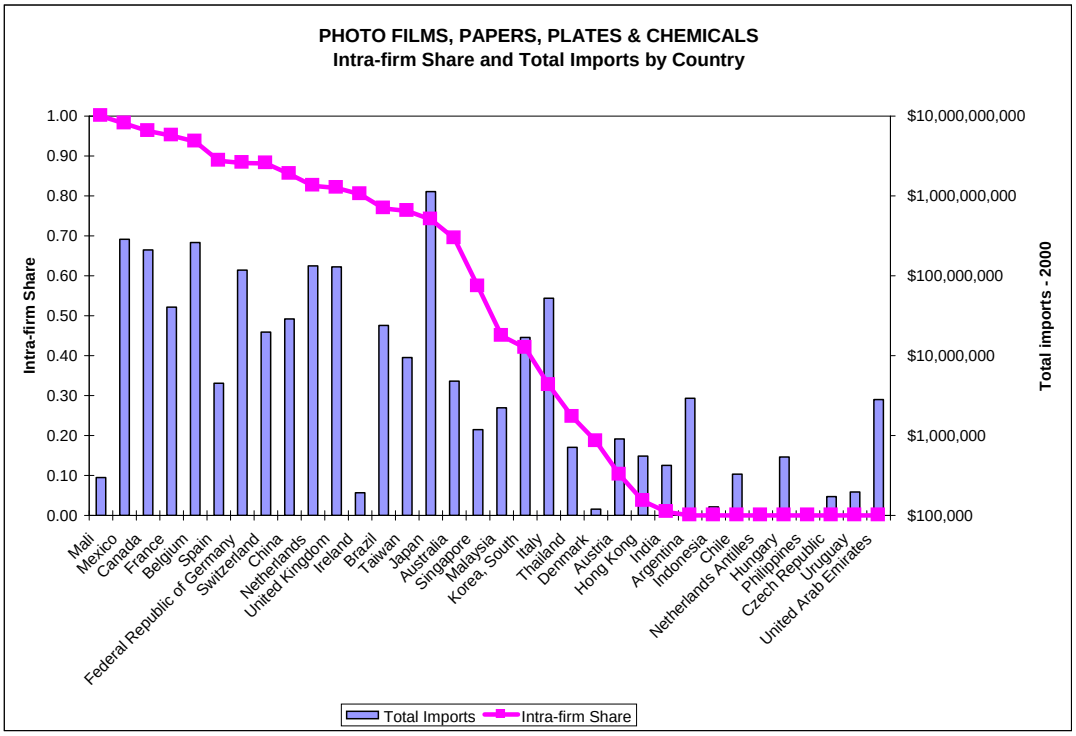


Figure 1: Intra-firm Import Share and Total Imports in 2000, NAICS Industry 325992

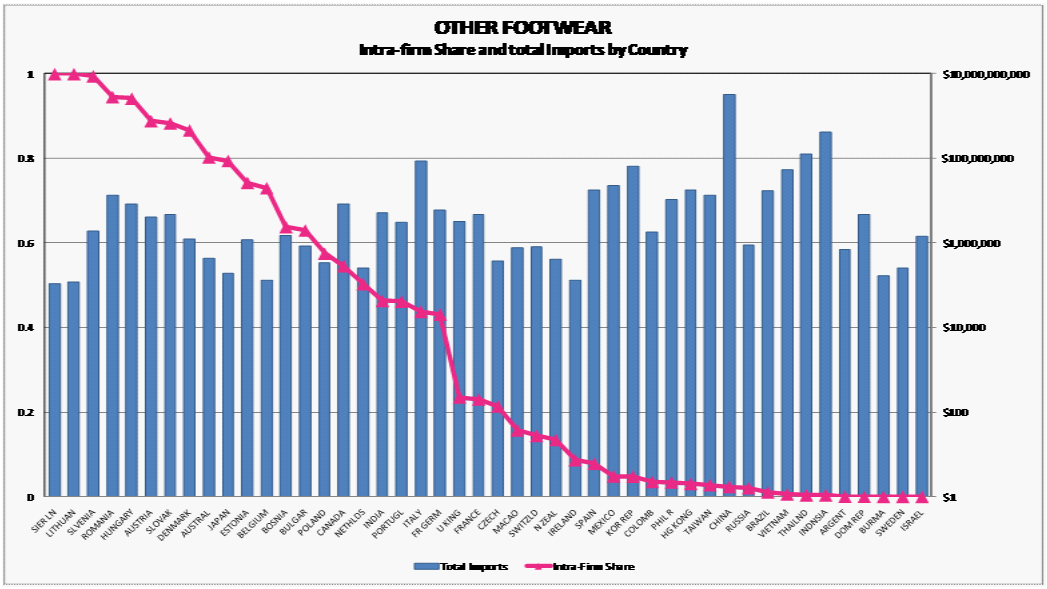


Figure 2: Intra-Firm Import Share and Total Imports in 2000, NAICS Industry 316219

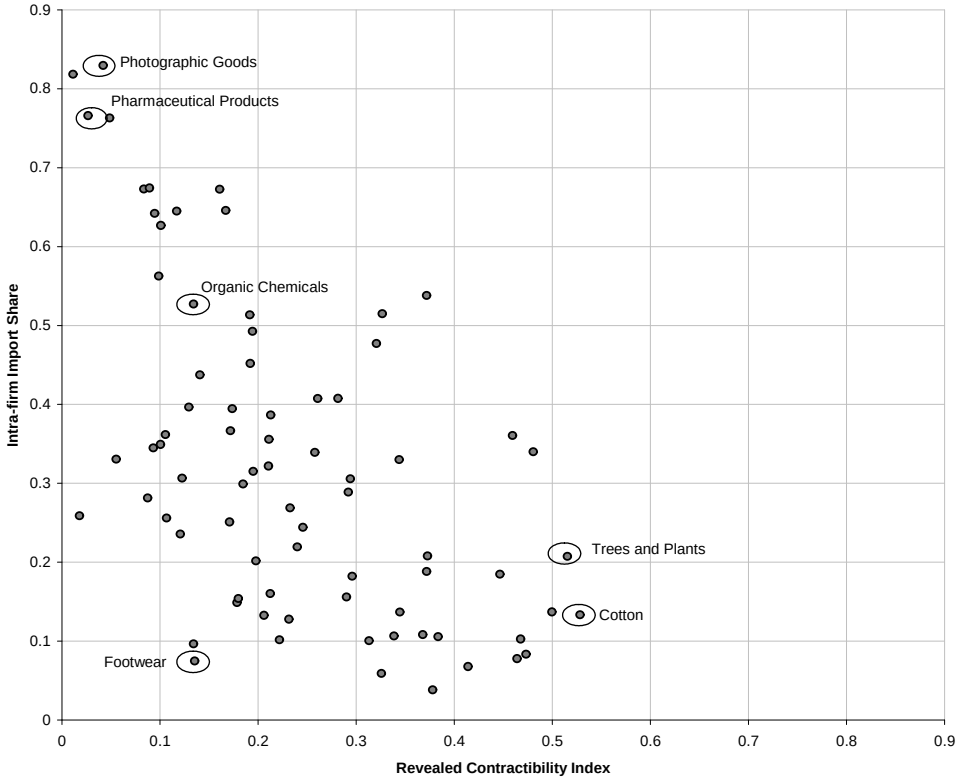


Figure 3: Intra-firm Import Intensity and "Revealed Contractability" by Two-Digit HS Category, 1997